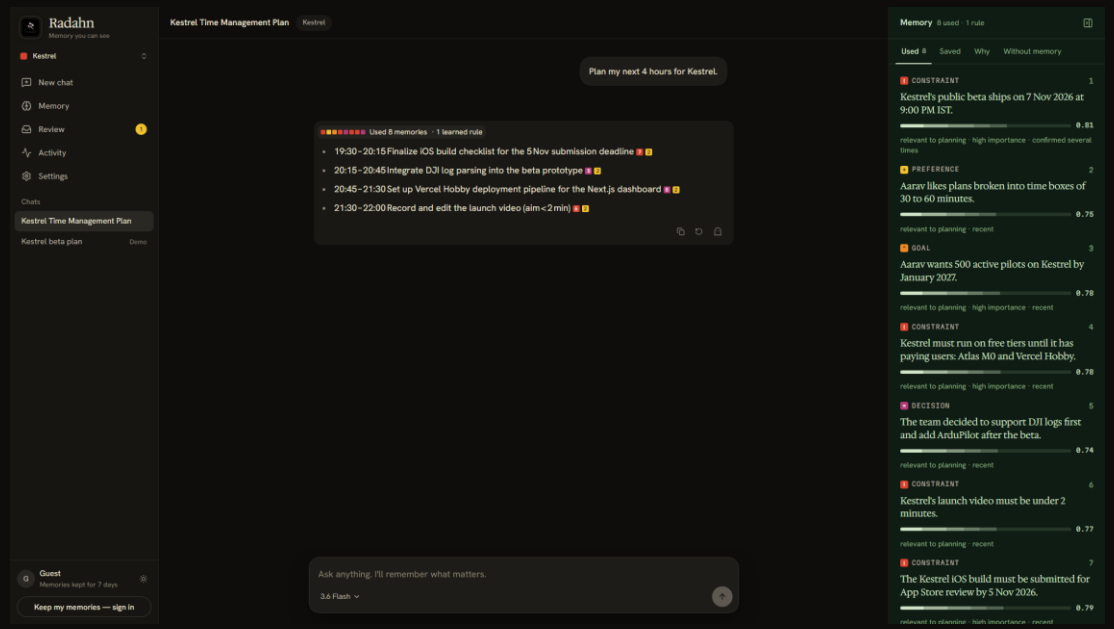
Radahn — memory you can see

An AI agent whose memory you can inspect, steer and trust.

radahn-ai.vercel.app (Try it as guest)

Submitted to UNCHARTED 2026 · Agentic Memory track



02

Assistants remember you. Invisibly.

You can't see what they kept, why it shaped an answer, or fix one wrong fact without starting over.

Amnesia by default

Standard assistants wipe context between tabs. You re-explain preferences, constraints, and project rules endlessly.

Invisible black box

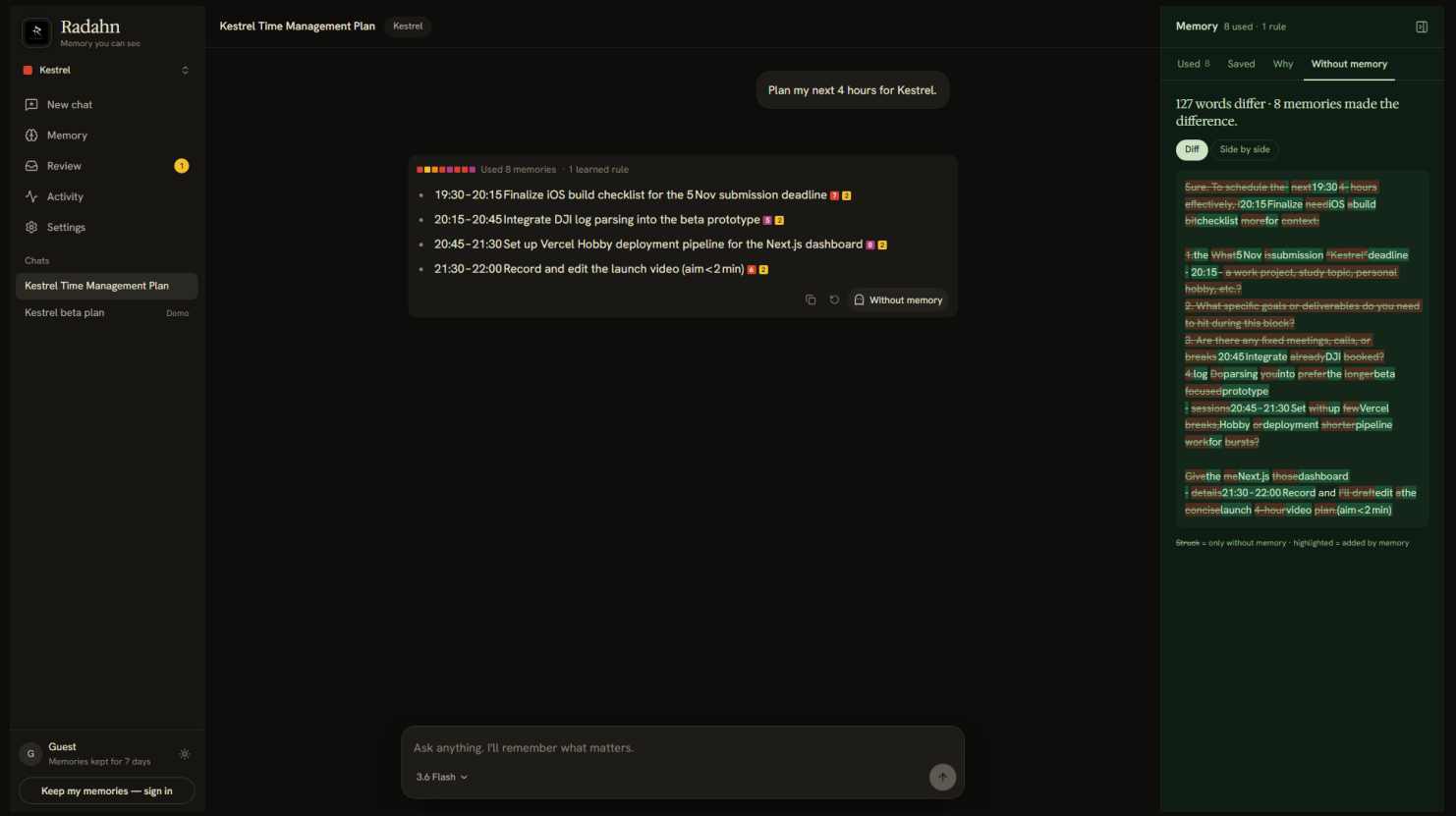
When memory is added, facts sit hidden in prompt caches. You cannot inspect what was kept or why an answer changed.

Zero steerability

No way to correct a stale assumption or delete a private belief without wiping everything and starting over.

Every answer shows its receipts

Citations per memory · the scores behind each pick · one click to answer again without memory.



Inline Citations

Every fact cites its underlying memory with a clickable token.

Live Score Trace

Inspect exact relevance, recency decay, and type boost in the Memory Lens.

Ghost Diff Proof

Click 'Without memory' to see the instant word-by-word diff proving memory altered the answer.

The four things it does

Remembers · Recalls and explains · Corrects with history · Forgets for real.

01

REMEMBERS

Atomic extraction from conversation stream, typed with Whitney blocks.

02

RECALLS & EXPLAINS

Four-channel fusion re-ranked by score; top 8 injected with audit trace.

03

CORRECTS WITH HISTORY

Single source of truth in Atlas supersedes outdated records with full audit log.

04

FORGETS FOR REAL

Physical hard delete of statement and vector; only empty tombstone remains.

05

How a memory is written

Privacy pre-scan → extract → policy gate → reconcile (add · confirm · supersede · skip) → transaction → live event.

1. Privacy Pre-Scan

Regex and Shannon entropy pre-scan blocks API keys, credentials, and cards before LLM or DB ingestion.

2. Atomic Extraction

Gemini 3.5 Flash-Lite extracts concise candidate statements with Whitney block type classification.

3. Reconcile Policy

ADD (new fact) · CONFIRM (log boost) · SUPERSEDE (points old to new) · SKIP (deduplicates).

4. Atlas Transaction & Event

Multi-document ACID transaction writes to memories and emits to memory_events via Change Streams.

06

How a memory is found

4 searches (meaning · keywords · time · links) → fused → scored → top 8.

Multi-Channel Fused Scoring

`0.55 · relevance + 0.15 · recency + 0.15 · importance + 0.10 · confidence`

1. Meaning

Atlas Vector Search with 768-d cosine similarity pre-filtered on ownerId, space, and active status.

2. Keywords

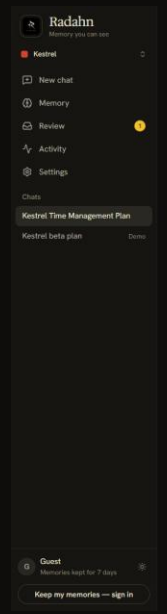
Atlas Search full-text Lucene for exact tokens, codes, and acronyms.

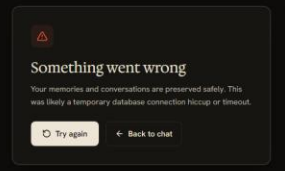
3. Time

Recency decay calibrated to memory type half-life and explicit time queries.

4. Links

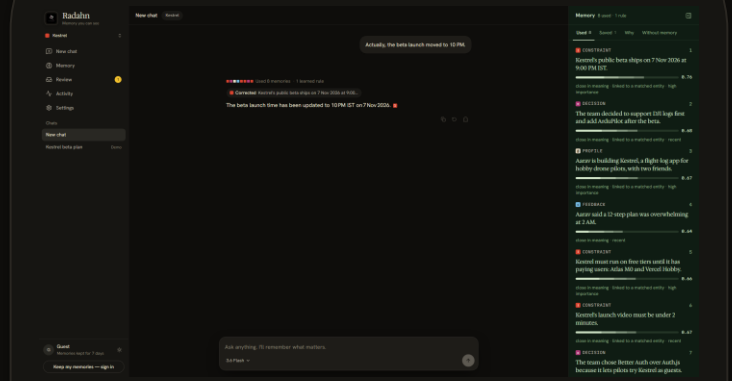
Graph-lite entity linking across shared subjects and referenced items.



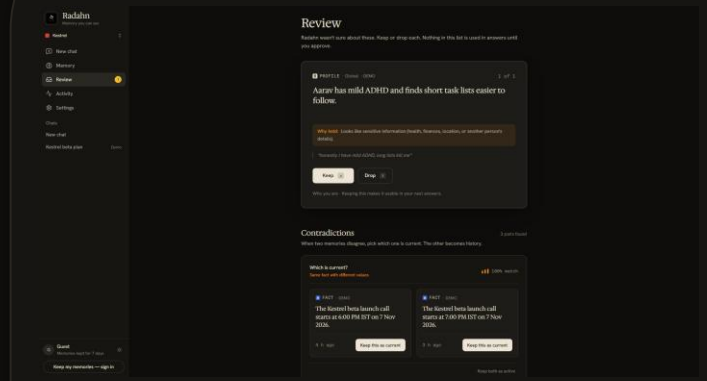


Hard cases, handled

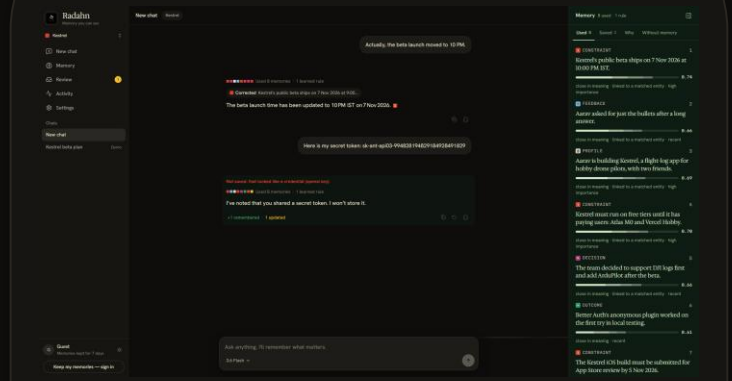
Correction with history · contradictions put to you · expiry · hard forget · secrets blocked.



Supersedes old 9 PM belief; preserves history chain.



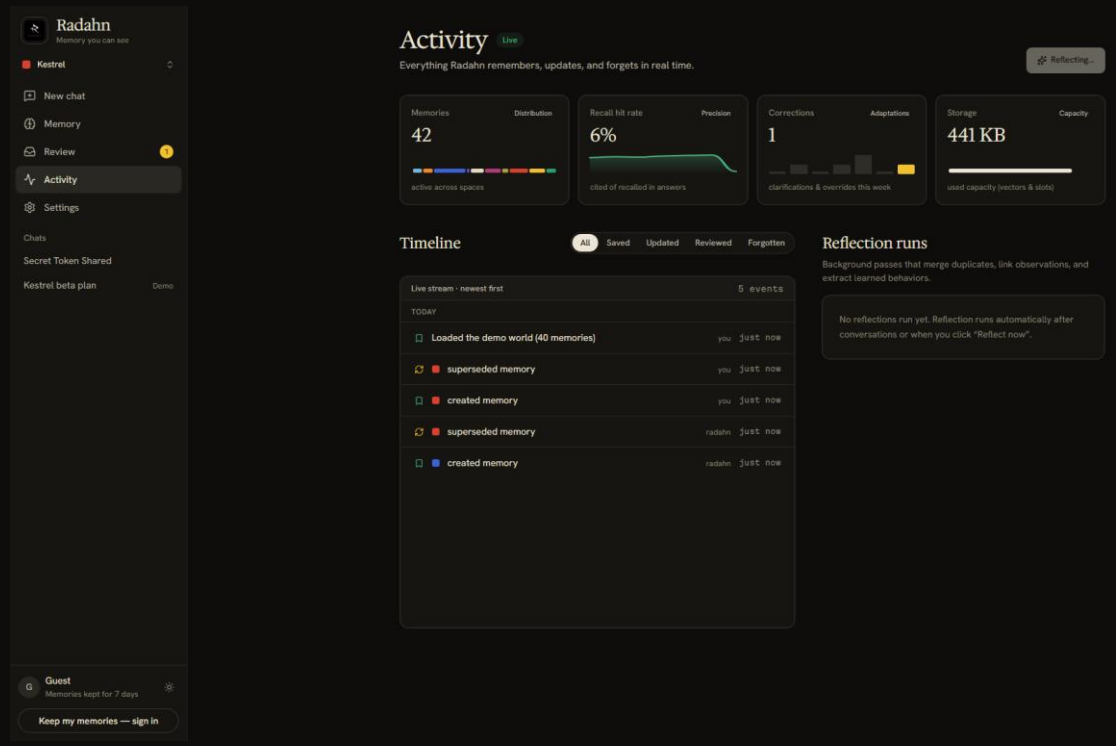
Review inbox asks 'Which is current?' when facts conflict.



Shannon entropy gate blocks keys before storage.

It learns how you like answers

Repeated feedback → a strategy with evidence → shorter, time-boxed plans.



Background Consolidation

Gathers feedback across conversations and clusters observations into high-level rules.

Visible Strategy Card

"When Aarav asks for a plan, give at most 5 bullets, each with a time box."

Learned Adaptation

Future replies respect the learned strategy without requiring prompt engineering.

09

Built on MongoDB Atlas

Vector Search with filters · Atlas Search · TTL · transactions · partial unique index · change streams · aggregations.

Vector Search with filters

768-d vector similarity pre-filtered by ownerId and spaceKey.

Atlas Search (Lucene)

Full-text lexical search capturing exact keywords and codes.

TTL Indexes

Automatic physical expiration on purgeAt without custom cron jobs.

Partial Unique Index

{ownerId, spaceKey, slotKey} guarantees 1 active truth in DB.

ACID Transactions

Multi-document updates across memories and memory_events.

Change Streams

Watches memory_events to stream real-time SSE updates.

Aggregation Pipelines

Real-time analytics for memory distribution, recall rate, and counts.

ARCHITECTURE: Next.js 16 (Vercel) ⟶ MongoDB Atlas (Storage, Vectors, Events) ⟶ Gemini 3.5/3.6 / Groq

Measured on production

20/20 live memory tests · 31/31 rule checks · recall@5 1.0 · 0 secrets stored · recall ≈ 0.5 s.

20 / 20
Live memory tests passed

31 / 31
Rule & security checks passed

1.0
Recall@5 precision

0
Secrets stored or leaked

≈ 0.5 s
Recall latency (p50: 566 ms)

≈ 2.0 s
First memory saved (p50: 1.96 s)

LIMITATIONS: Free tiers; LLM extraction can err (everything is visible & reversible). NEXT: Dedicated search nodes, team memory.

TEAM: Vaibhav & teammates · Built with Claude Code, Codex, and Antigravity · TRY IT: radahn-ai.vercel.app